

Problem Sheet: Trig Substitutions 1 Solution

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Exercise 1. Using a trigonometric substitution, find the indefinite integral

$$\int \sqrt{1 - 4x^2} \, dx$$

Then state the interval of x values where this substitution is valid.

Solution. The identities that look like $1 - 4x^2$ are

$$1 - \sin^2 \theta = \cos^2 \theta, \quad 1 - \cos^2 \theta = \sin^2 \theta$$

so we use a scaled version of the first and let

$$2x = \sin \theta$$

with

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

This is a convenient choice because then $\cos \theta \geq 0$.

Therefore

$$x = \frac{1}{2} \sin \theta$$

and

$$\frac{dx}{d\theta} = \frac{1}{2} \cos \theta$$

so

$$dx = \frac{1}{2} \cos \theta \, d\theta.$$

Now substitute from left to right:

$$\begin{aligned} \int \sqrt{1 - 4x^2} \, dx &= \int \sqrt{1 - \sin^2 \theta} \, dx && \text{(using } 2x = \sin \theta) \\ &= \int \sqrt{\cos^2 \theta} \cdot \frac{1}{2} \cos \theta \, d\theta && \text{(using } dx = \frac{1}{2} \cos \theta \, d\theta) \\ &= \frac{1}{2} \int \cos^2 \theta \, d\theta \end{aligned}$$

Here we used that

$$\sqrt{\cos^2 \theta} = \cos \theta,$$

which is valid on our chosen interval because $\cos \theta \geq 0$ there.

Now use

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

to find

$$\begin{aligned}\frac{1}{2} \int \cos^2 \theta \, d\theta &= \frac{1}{2} \int \frac{1 + \cos 2\theta}{2} \, d\theta \\ &= \frac{1}{4} \int 1 \, d\theta + \frac{1}{4} \int \cos 2\theta \, d\theta \\ &= \frac{1}{4} \theta + \frac{1}{8} \sin 2\theta + c\end{aligned}$$

Now return to x . Since $2x = \sin \theta$, we have

$$\theta = \sin^{-1}(2x)$$

and also

$$\sin 2\theta = 2 \sin \theta \cos \theta.$$

Therefore

$$\begin{aligned}\frac{1}{4} \theta + \frac{1}{8} \sin 2\theta + c &= \frac{1}{4} \theta + \frac{1}{8} (2 \sin \theta \cos \theta) + c \\ &= \frac{1}{4} \theta + \frac{1}{4} \sin \theta \cos \theta + c \\ &= \frac{1}{4} \theta + \frac{1}{4} (2x) \sqrt{1 - \sin^2 \theta} + c \\ &= \frac{1}{4} \theta + \frac{1}{2} x \sqrt{1 - 4x^2} + c \\ &= \frac{1}{4} \sin^{-1}(2x) + \frac{1}{2} x \sqrt{1 - 4x^2} + c\end{aligned}$$

Hence

$$\int \sqrt{1 - 4x^2} \, dx = \frac{1}{2} x \sqrt{1 - 4x^2} + \frac{1}{4} \sin^{-1}(2x) + c$$

When using this substitution, we are working in the range

$$-1 \leq 2x \leq 1$$

which gives

$$-\frac{1}{2} \leq x \leq \frac{1}{2}.$$

This is also exactly the range where the square root term $\sqrt{1 - 4x^2}$ is defined.

□

Exercise 2. Using a trigonometric substitution, compute

$$\int_{\frac{1}{2}}^1 \frac{1}{x \sqrt{1 - x^2}} \, dx.$$

Solution. Since the integrand contains $\sqrt{1 - x^2}$, we use

$$x = \cos \theta.$$

Since the bounds satisfy $\frac{1}{2} \leq x \leq 1$, we take

$$0 \leq \theta \leq \frac{\pi}{2},$$

so that both $\cos \theta \geq 0$ and $\sin \theta \geq 0$.

Then

$$\begin{aligned}x &= \cos \theta \\dx &= -\sin \theta \, d\theta \\ \sqrt{1-x^2} &= \sqrt{1-\cos^2 \theta} = \sin \theta\end{aligned}$$

where the final step is valid because $\sin \theta \geq 0$ on the interval we use.

Now convert the bounds:

$$\begin{aligned}x = \frac{1}{2} &\therefore \cos \theta = \frac{1}{2} \therefore \theta = \frac{\pi}{3} \\x = 1 &\therefore \cos \theta = 1 \therefore \theta = 0\end{aligned}$$

Substitute from left to right:

$$\begin{aligned}\int_{\frac{1}{2}}^1 \frac{1}{x\sqrt{1-x^2}} \, dx &= \int_{x=\frac{1}{2}}^{x=1} \frac{1}{x\sqrt{1-x^2}} \, dx \\&= \int_{\theta=\frac{\pi}{3}}^{\theta=0} \frac{1}{\cos \theta \sqrt{1-\cos^2 \theta}} (-\sin \theta) \, d\theta \\&= \int_{\theta=\frac{\pi}{3}}^{\theta=0} \frac{-\sin \theta}{\cos \theta \sin \theta} \, d\theta \\&= \int_{\theta=\frac{\pi}{3}}^{\theta=0} -\sec \theta \, d\theta \\&= \int_0^{\frac{\pi}{3}} \sec \theta \, d\theta\end{aligned}$$

Now use the formula book:

$$\int \sec \theta \, d\theta = \ln |\sec \theta + \tan \theta| + c$$

Therefore

$$\begin{aligned}\int_0^{\frac{\pi}{3}} \sec \theta \, d\theta &= [\ln |\sec \theta + \tan \theta|]_{\theta=0}^{\theta=\frac{\pi}{3}} \\&= \ln \left(\sec \frac{\pi}{3} + \tan \frac{\pi}{3} \right) - \ln(\sec 0 + \tan 0)\end{aligned}$$

Evaluate the trig values:

$$\sec \frac{\pi}{3} = 2, \quad \tan \frac{\pi}{3} = \sqrt{3}, \quad \sec 0 = 1, \quad \tan 0 = 0.$$

So

$$\begin{aligned}\int_{\frac{1}{2}}^1 \frac{1}{x\sqrt{1-x^2}} \, dx &= \ln(2 + \sqrt{3}) - \ln(1) \\&= \ln(2 + \sqrt{3})\end{aligned}$$

□

Exercise 3. Compute the following indefinite integral, assuming that $4 - x^2 \geq 0$ and $x \neq 0$.

$$\int \frac{1}{x^2 \sqrt{4-x^2}} \, dx$$

Solution. We look for a trig identity that matches $4 - x^2$. Since

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \\ 4 \sin^2 \theta + 4 \cos^2 \theta &= 4 \\ 4 \sin^2 \theta &= 4 - 4 \cos^2 \theta\end{aligned}$$

we choose

$$x = 2 \cos \theta.$$

We take

$$0 \leq \theta \leq \pi$$

so that $\sin \theta \geq 0$, which will make the square root step below valid.

Then

$$\begin{aligned}x^2 &= 4 \cos^2 \theta \\ \sqrt{4 - x^2} &= \sqrt{4 - 4 \cos^2 \theta} = 2 \sin \theta \\ \frac{dx}{d\theta} &= -2 \sin \theta \\ dx &= -2 \sin \theta d\theta\end{aligned}$$

Now substitute from left to right:

$$\begin{aligned}\int \frac{1}{x^2 \sqrt{4 - x^2}} dx &= \int \frac{1}{4 \cos^2 \theta \sqrt{4 - x^2}} dx && \text{(using } x^2 = 4 \cos^2 \theta \text{)} \\ &= \int \frac{1}{4 \cos^2 \theta \cdot 2 \sin \theta} dx && \text{(using } \sqrt{4 - x^2} = 2 \sin \theta \text{)} \\ &= \int \frac{1}{4 \cos^2 \theta \cdot 2 \sin \theta} (-2 \sin \theta) d\theta && \text{(using } dx = -2 \sin \theta d\theta \text{)} \\ &= -\frac{1}{4} \int \sec^2 \theta d\theta\end{aligned}$$

Using the formula book,

$$\int \sec^2 \theta d\theta = \tan \theta + c$$

so

$$\int \frac{1}{x^2 \sqrt{4 - x^2}} dx = -\frac{1}{4} \tan \theta + c.$$

Now write everything back in terms of x . Since $x = 2 \cos \theta$, we have

$$\cos \theta = \frac{x}{2}$$

and therefore

$$\begin{aligned}\tan \theta &= \frac{\sin \theta}{\cos \theta} \\ &= \frac{\sqrt{1 - \cos^2 \theta}}{\cos \theta} \\ &= \frac{\sqrt{1 - \left(\frac{x}{2}\right)^2}}{\frac{x}{2}} \\ &= \frac{\sqrt{\frac{4 - x^2}{4}}}{\frac{x}{2}} \\ &= \frac{\frac{1}{2} \sqrt{4 - x^2}}{\frac{x}{2}} \\ &= \frac{\sqrt{4 - x^2}}{x}\end{aligned}$$

Hence

$$\int \frac{1}{x^2 \sqrt{4-x^2}} dx = -\frac{1}{4} \cdot \frac{\sqrt{4-x^2}}{x} + c$$

that is,

$$\int \frac{1}{x^2 \sqrt{4-x^2}} dx = -\frac{\sqrt{4-x^2}}{4x} + c.$$

Since the original integrand contains x^2 in the denominator, we also require

$$x \neq 0.$$

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